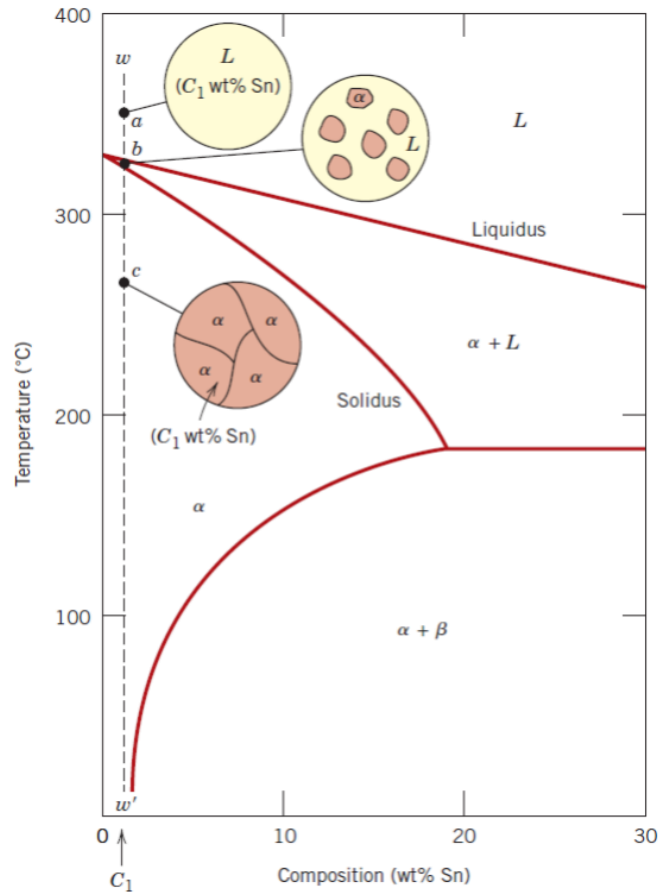


TOPIC 4. Phase Diagrams II

- **Precipitation in solid state.**
- **Invariant reactions.**
- **Intermetallic compounds.**
- **Congruent and incongruent melting.**

SOLID STATE PRECIPITATION

Microstructure below the solid solubility limit



a) 350°C, 1% Sn
[liquid]

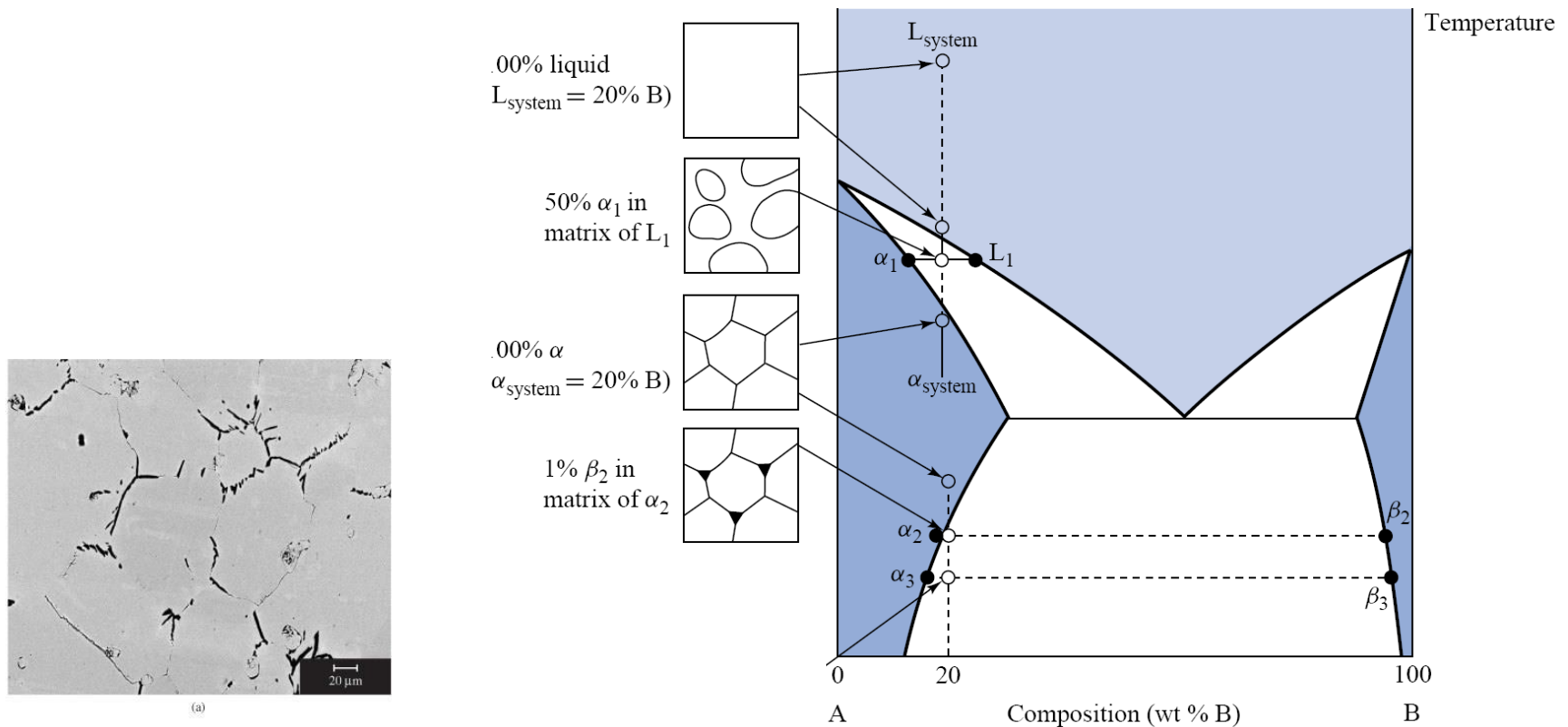
b) 330°C, Solid phase α begins to form
[$\alpha + L$]

c) 270°C, 1% Sn
[Phase α]

SOLID STATE PRECIPITATION

Microstructure above solid solubility limit: solid state precipitation

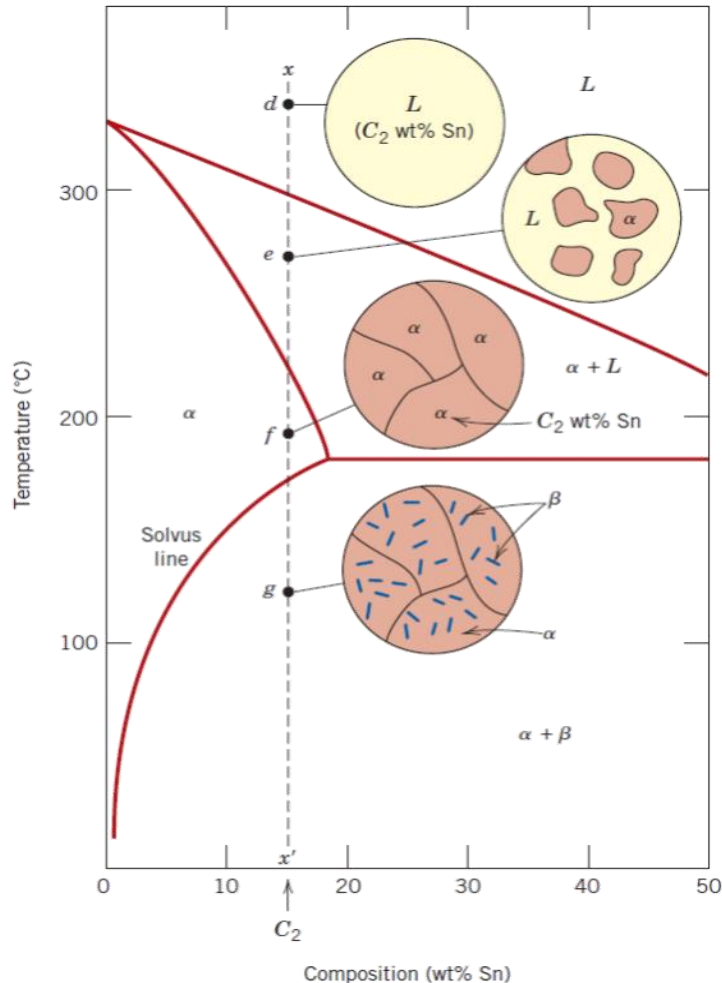
Precipitation in solid state : When crossing the solvus line, the α solid solubility is exceeded $\rightarrow$ formation of small β -phase particles, which grow as $\downarrow T$.



SOLID STATE PRECIPITATION

Microstructure above solid solubility limit: solid state precipitation

Precipitation in solid state: Example



d) 350°C, 16% Sn, [**Liquid**]

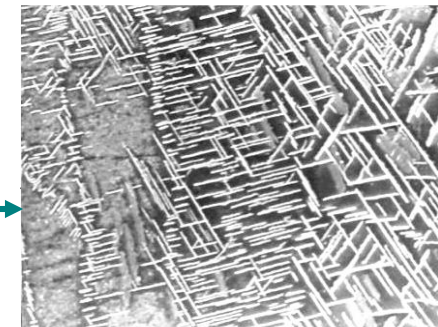
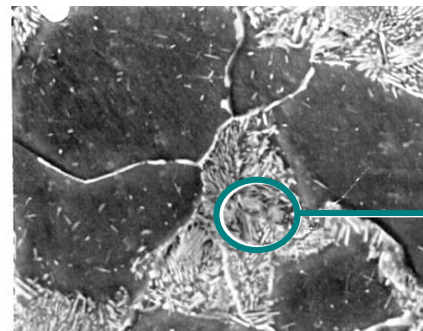
e) 270°C. Solid phase α begins to form, [**$\alpha + L$**]

f) 192°C, 16% Sn, [**Phase α**]

g) 121°C, [**Phases $\alpha + \beta$**]

Precipitation in solid state

When crossing the solvus line, the α solid solubility is exceeded $\rightarrow$ formation of small β -phase particles, which grow as $\downarrow T$.

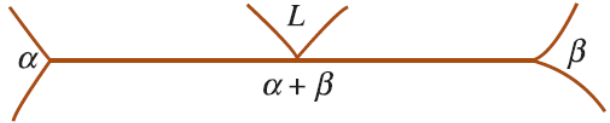
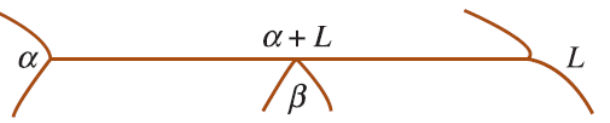
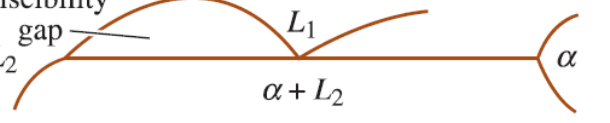
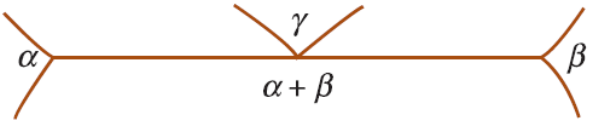
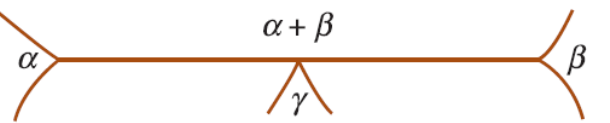


TOPIC 4. Phase Diagrams II

- Precipitation in solid state.
- **Invariant reactions.**
- Intermetallic compounds.
- Congruent and incongruent melting.

INVARIANT REACTIONS

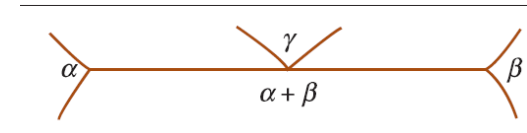
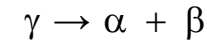
solid state
reactions

Eutectic	$L \rightarrow \alpha + \beta$	
Peritectic	$\alpha + L \rightarrow \beta$	
Monotectic	$L_1 \rightarrow L_2 + \alpha$	
Eutectoid	$\gamma \rightarrow \alpha + \beta$	
Peritectoid	$\alpha + \beta \rightarrow \gamma$	

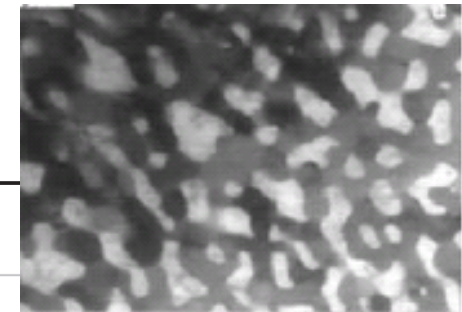
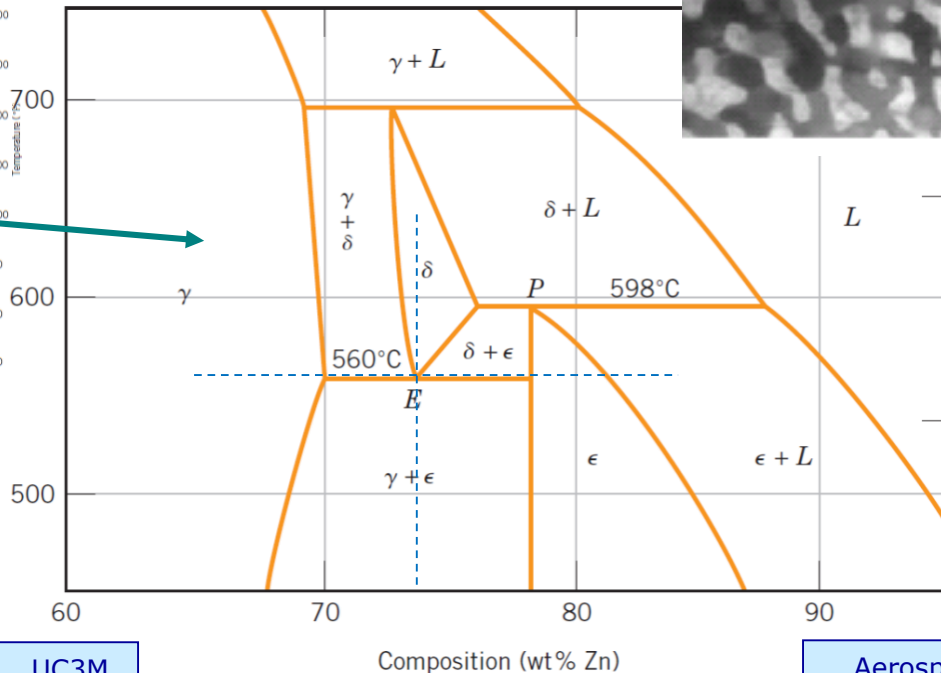
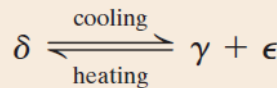
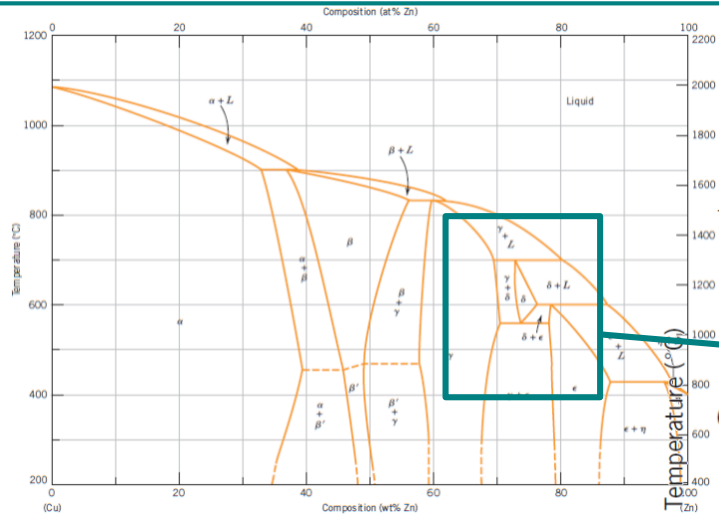
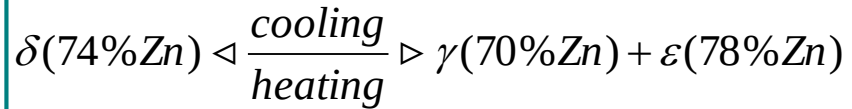
INVARIANT REACTIONS

Eutectoid reaction: invariant reaction where a solid phase transforms into two solid phases.

Eutectoid reaction:



Example: Cu-Zn at 560°C (74% Zn)

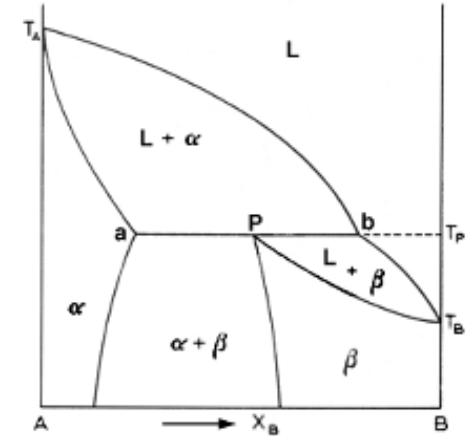


INVARIANT REACTIONS

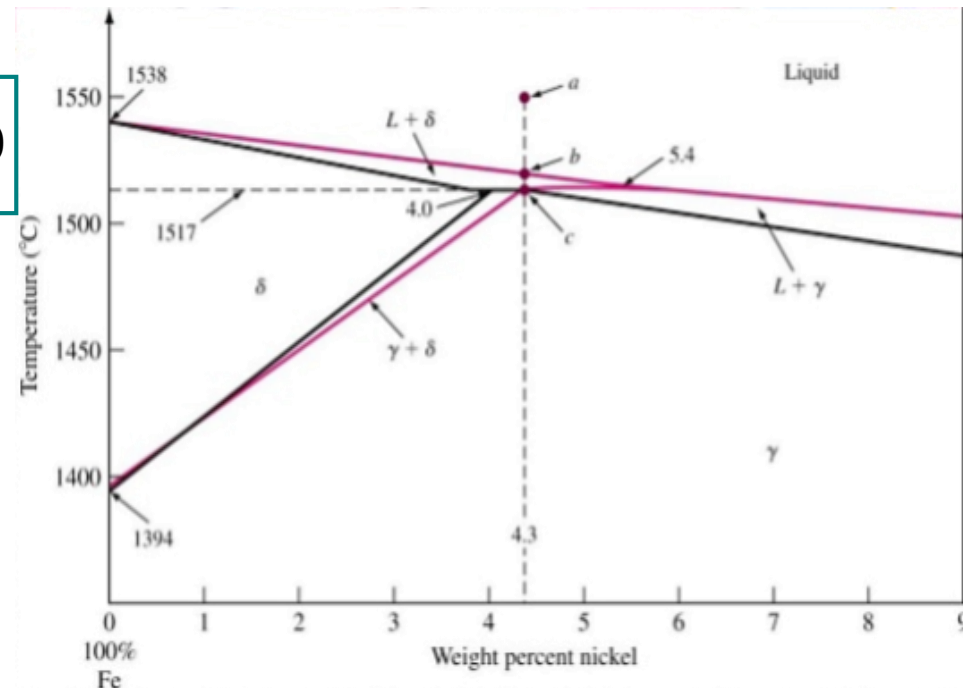
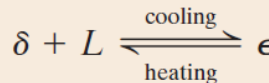
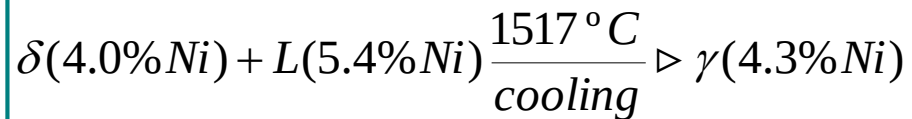
Peritectic reaction: invariant reaction where a solid and a liquid phase transform into a different solid phase

Peritectic: invariant **point** where a solid and a liquid phase transform in a different solid phase.

Peritectic reaction:



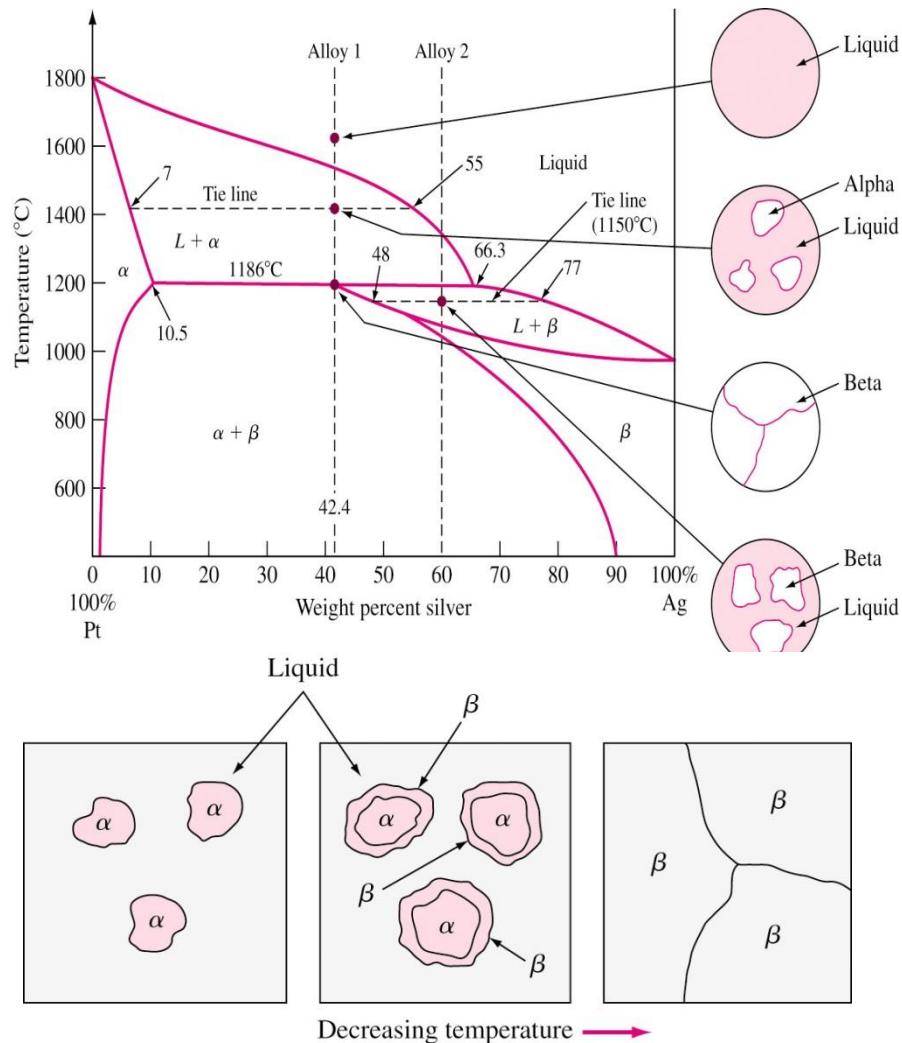
Example: Fe-Ni at 1517°C and 4.3% Ni



INVARIANT REACTIONS

Peritectic reaction

Example: Pt-Ag at 1186°C (42.4% Ag)



- At 42.4 % Ag and 1400°C**

Phases present: Liquid Alpha

Composition: 55% Ag 7%Ag

Amount of Phases:

Liquid: $(42.4 - 7) / (55 - 7) = 74\%$

Alpha: $(55 - 42.4) / (55 - 7) = 26\%$
- At 42.4% Ag and 1186°C - ΔT**

Phase Present: Beta only

Composition: 42.4% Ag

Amount of Phase: 100%
- At 42.4% Ag and 1186°C + ΔT**

Phases present: Liquid Alpha

Composition: 66.3% Ag 10.5%Ag

Amount of Phases:

Liquid: $(42.4 - 10.5) / (66.3 - 10.5) = 57\%$

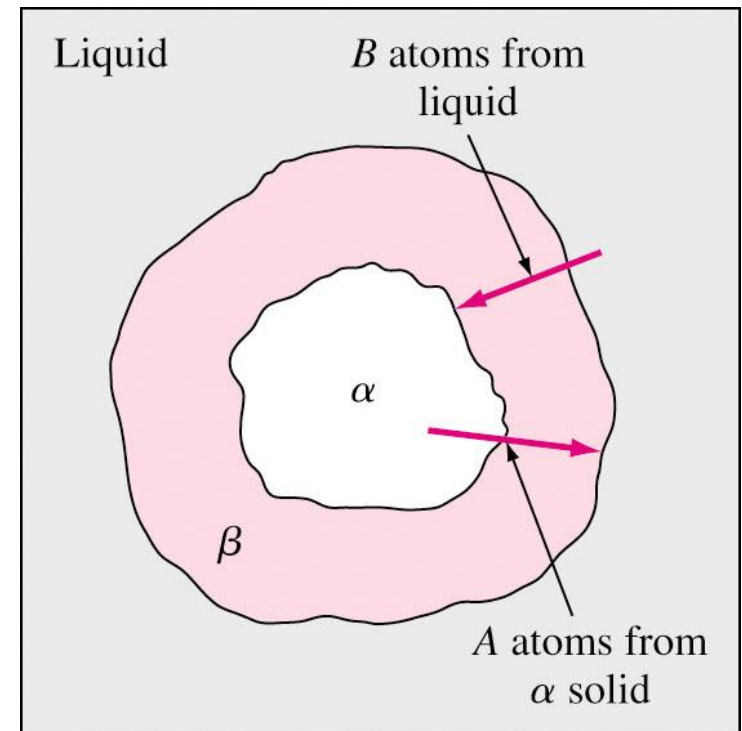
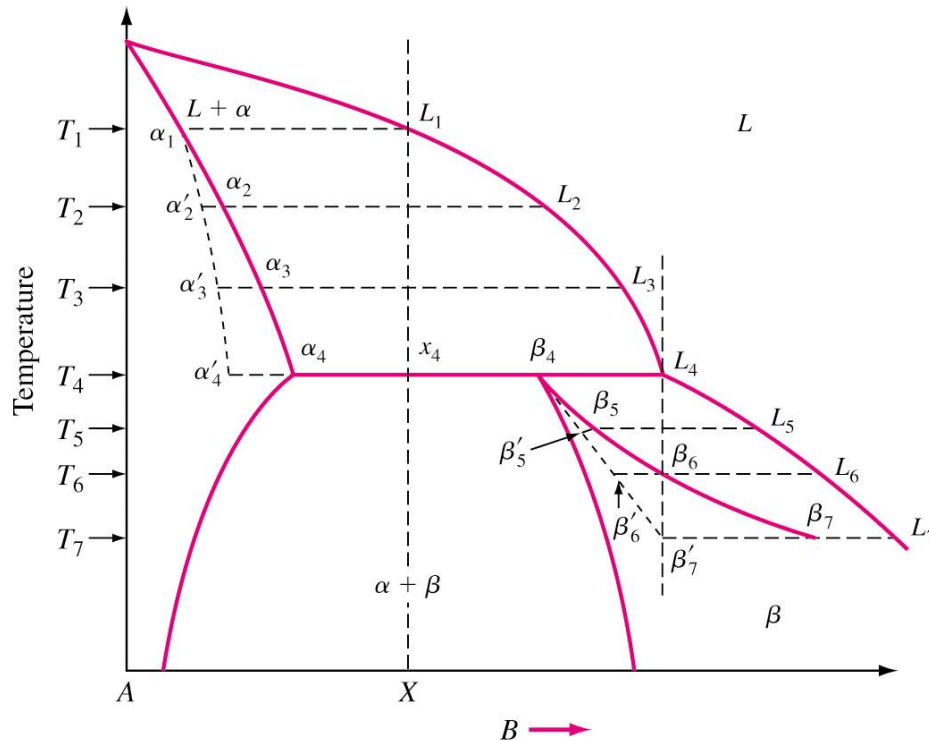
Alpha: $(66.3 - 10.5) / (66.3 - 10.5) = 43\%$

INVARIANT REACTIONS

Peritectic reaction

Rapid (non-equilibrium) Solidification in Peritectic System

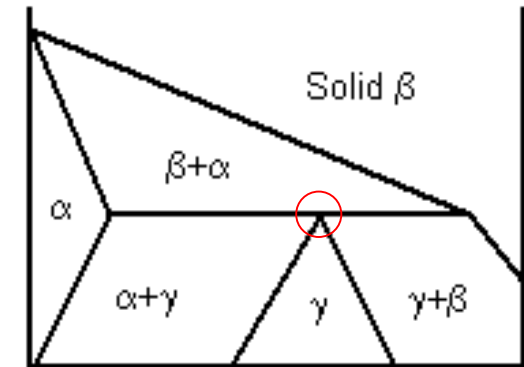
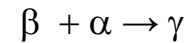
- **Surrounding or Encasement:** During peritectic reaction, $L + \alpha \rightarrow \beta$, the beta phase created surrounds primary alpha.
- Beta creates diffusion barrier resulting in coring.



INVARIANT REACTIONS

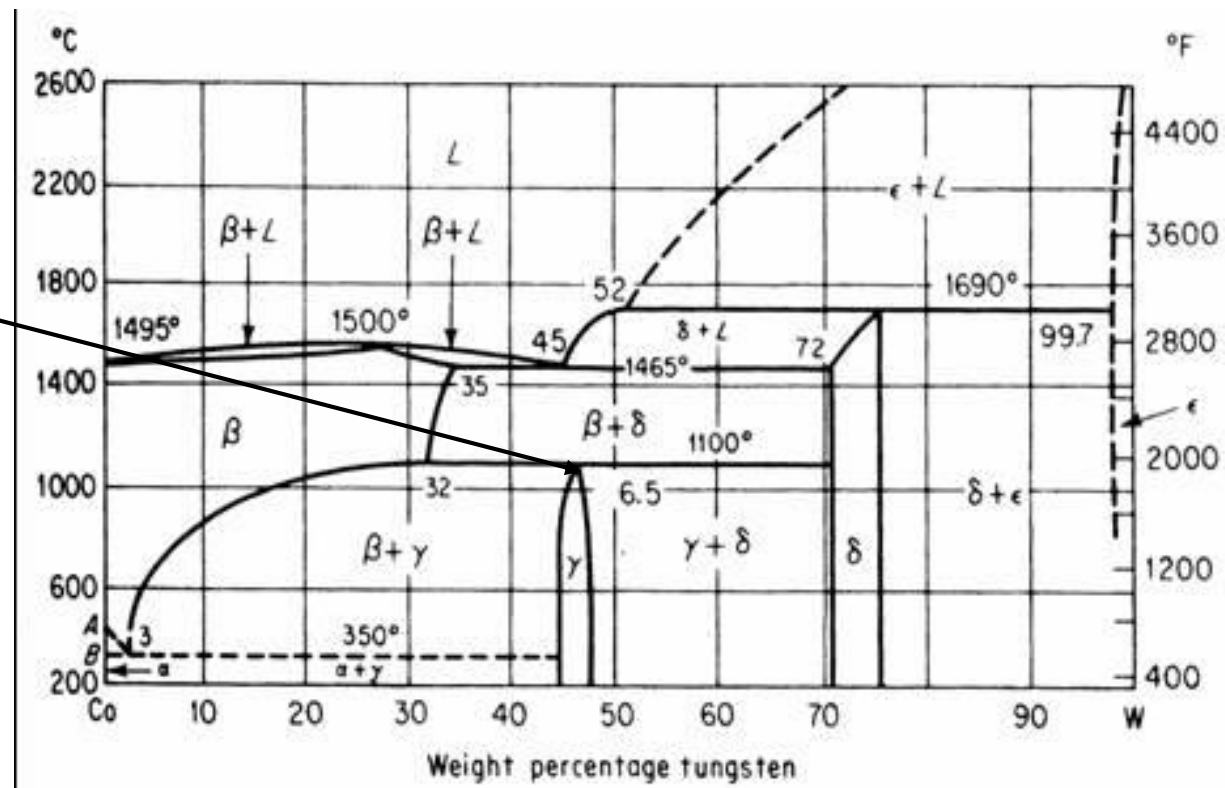
Peritectoid reaction: invariant reaction where two solid phases transform into a new solid phase

Peritectoid reaction:



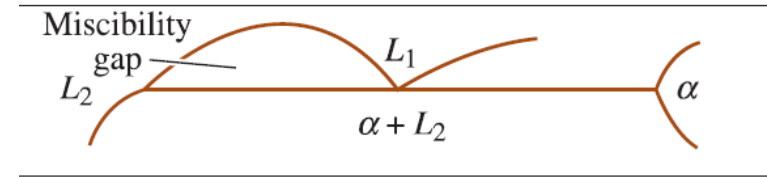
Peritectoid: invariant point where two solid phases transform into a new solid phase.

PERITECTOID



INVARIANT REACTIONS

Monotectic reaction: invariant reaction where a liquid phase transforms into a solid phase + another liquid one (*liquid miscibility gap*)

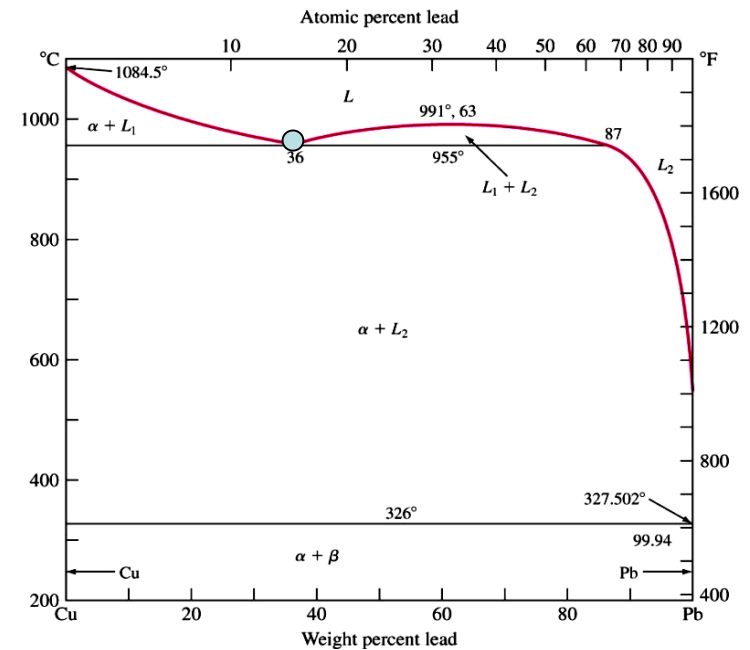
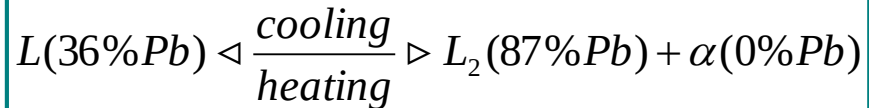


Monotectic: invariant point where a liquid phase transforms into a solid phase + another liquid one.

Monotectic reaction:



Example: Cu-Pb at 955°C (36%Pb)



TOPIC 4. Phase Diagrams II

- Precipitation in solid state.
- Invariant reactions.
- **Intermetallic compounds.**
- Congruent and incongruent melting.

INTERMETALLIC COMPOUNDS

Intermetallic compounds

It is made up of two or more elements, producing a new phase with its own composition, crystal structure and properties

Terminal phases → occur at the ends of phase diagrams

Intermediate phases → occur in a composition range inside the phase diagram

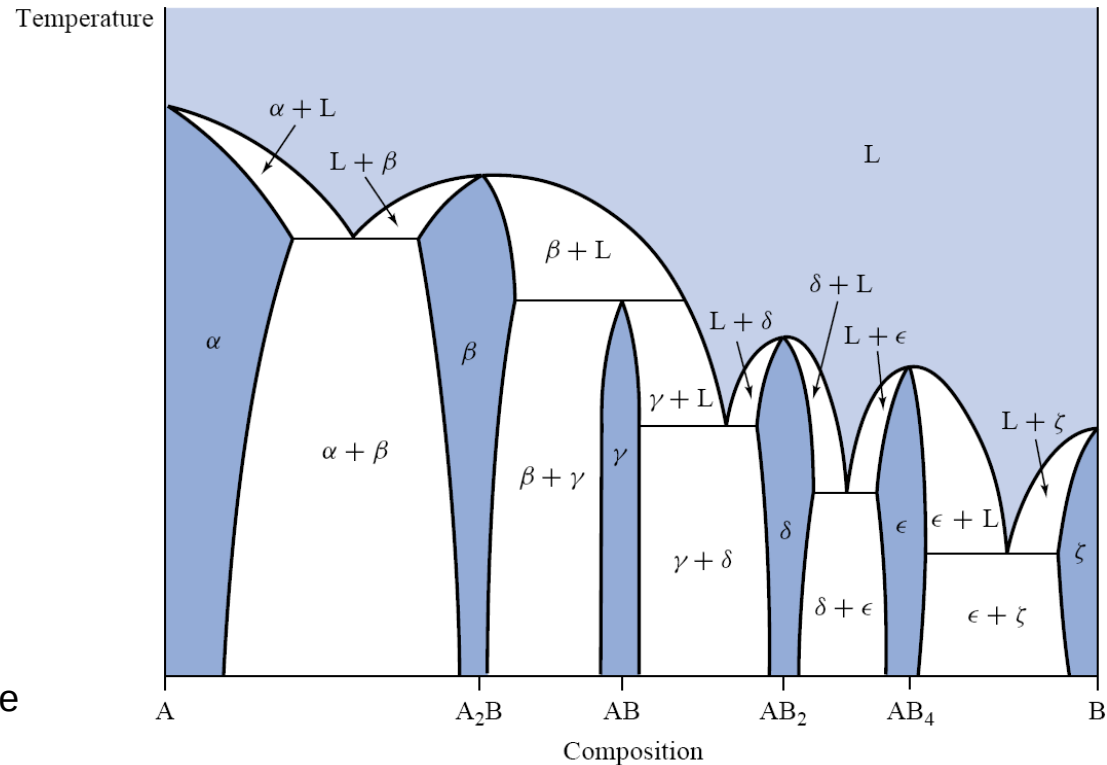
Terminal phases:

α , ξ

Intermediate phases:

β , γ , δ , ϵ

- Percent Ionic/Covalent bond of the intermetallic compound depends on electronegativity difference between the elements



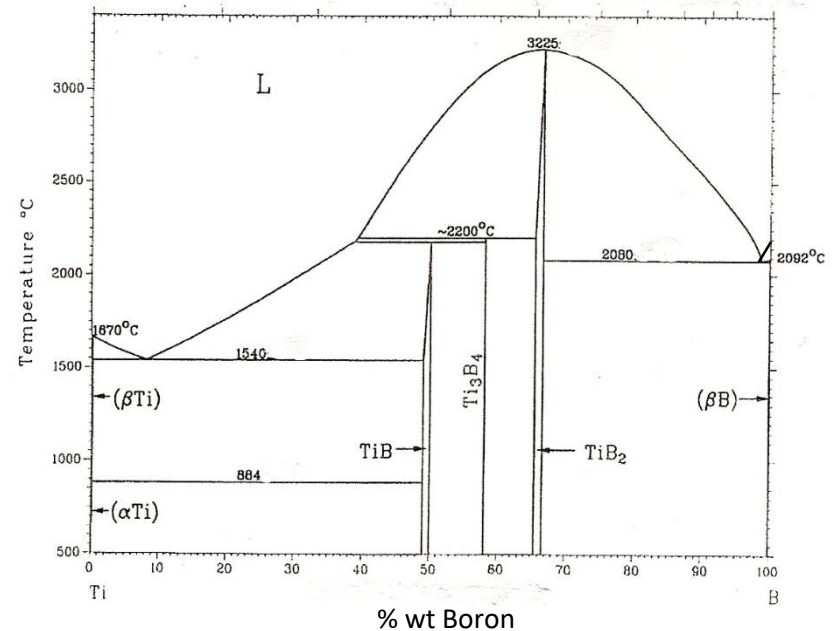
INTERMETALLIC COMPOUNDS

Intermetallic compounds

It is made up of two or more components, producing a new phase with its own composition, crystal structure and properties.

- **Stoichiometric** intermetallic compound: fixed composition/ratio of involved atoms → represented by a vertical line in the phase diagram
- **Non- stoichiometric** intermetallic compound: range of compositions (intermediate solid solution).

- Stoichiometric: Ti_3B_4
- Non-stoichiometric : TiB , TiB_2



TOPIC 4. Phase Diagrams II

- Precipitation in solid state.
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- Intermetallic compounds.
- **Congruent and incongruent melting.**

CONGRUENT AND INCONGRUENT MELTING

Congruently melting: An intermetallic compound is said to **melt congruently** when the liquid formed after melting (solid $\rightarrow$ liquid transformation) maintains its composition e.g. MgNi_2 , pure metals

Incongruently melting: A intermetallic compound is said to **melt incongruently** when the composition is not maintained after melting (solid $\rightarrow$ liquid transformation) and produces a different solid upon cooling. e.g. Mg_2Ni , peritectic reaction: $\alpha + L \rightarrow \beta$..

